MedAgent Clinical Decision Support

Case ID: demo_case_5373f1 | **Status:** APPROVE | **Physician:** Dr. Sarah Chen, MD | **Date:** 2026-10-13 17:26 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: 64, Sex: male
Notes: 64yo male with 4 days of fever, purulent cough, right pleuritic chest pain and dyspnea.

Clinical Summary / Impression

Patient clinical evaluation demonstrates features consistent with Community-Acquired Pneumonia. Diagnostic synthesis is supported by key multimodal findings including fever, cough, pleuritic. Differential considerations include Sepsis / Severe Systemic Infection, Pleural Effusion. Targeted diagnostic workup and serial clinical monitoring are recommended to confirm etiology and guide definitive management.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	Community-Acquired Pneumonia	High	fever, cough, pleuritic, dyspnea, Pneumonia (p=52.5%)	S2, S3
2	Sepsis / Severe Systemic Infection	Moderate	Leukocytosis with Neutrophilia, Raised Inflammatory Markers	S_Sepsis_1, S_Seps
3	Pleural Effusion	Moderate	pleuritic, dyspnea, Raised Inflammatory Markers	S5, S8
4	COPD Acute Exacerbation	Moderate	dyspnea, Leukocytosis with Neutrophilia	S_COPDAc_1, S_C
5	Normocytic Anemia of Chronic Disease	Moderate	Raised Inflammatory Markers	S_Normoc_1, S_No

Radiographic Findings (Chest X-Ray)

Pathology	Probability	Location	Indicative Severity	Confidence
Enlarged Cardiomediastinum	52.7%	cardiac / mediastinal silhouette	Low	Moderate
Pneumonia	52.5%	bilateral lower zone	Moderate	Moderate
Emphysema	51.1%	left mid zone	Moderate	Moderate
Fracture	50.9%	bilateral lower zone	Low	Moderate
Lung Opacity	38.5%	bilateral mid zone	Moderate	Low

Laboratory Findings & Identified Patterns

Identified Clinical Patterns: Leukocytosis with Neutrophilia (soon); **Raised Inflammatory Markers** (routine)

Physician Attestation: I, Dr. Dr. Sarah Chen, MD, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
Reviewer Notes: Clinical differential and radiographic findings verified. Consistent with bacterial CAP.
Signature: _____ **Date:** 2026-10-13